TWO CONJECTURES OF SUN ON THE RECURRENCE $a(n+1) = 2n(n+1)a(n) - n^4a(n-1)$ ARE FALSE

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ABSTRACT. Let $a(0) = 1$, $a(1) = 0$ and $a(n+1) = 2n(n+1)a(n) - n^4a(n-1)$ (OEIS A217703). Z.-W. Sun conjectured that (ii) $a(n) < 0$ exactly when $1 < n < 58$ or $n > 2177$, and that (iii) $|a(n)|^{1/n} = o(n^2)$. Both are false. An exact computation shows $a(81935) > 0$. We also prove that $\limsup_n |a(n)|^{1/n}/n^2 = e^{-2}$, using the Casoratian of the recurrence and a polynomial bound for a second solution. The disproof of (iii) is formalized in Lean 4 with Mathlib, against the statement `OeisA217703.conjecture3` in Google DeepMind's Formal Conjectures repository.

1. INTRODUCTION

The sequence $a(n) = 1, 0, -1, -12, -207, \ldots$ is A217703 [2]. It is the constant term $S_n(0)$ of polynomials $S_n(x)$ that Sun introduced in 2013. In the OEIS entry Sun conjectured, among other things:

(ii) $a(n) < 0$ if and only if $1 < n < 58$ or $n > 2177$;
(iii) $|a(n)|^{1/n} = o(n^2)$ as $n \to \infty$.

They are stated in [1] as `OeisA217703.conjecture2` and `conjecture3`.

Theorem 1. (a) *$a(81935) > 0$; hence (ii) is false.*
(b) *$\limsup_{n\to\infty} |a(n)|^{1/n}/n^2 = e^{-2}$; hence (iii) is false.*

2. PROOF OF (A)

We evaluated the recurrence exactly with arbitrary-precision integers. The signs of $a(n)$ change at $n = 2, 58, 2178, 81935$; the next change, located with 40-digit floating point, is at 3082582. In particular $a(81935)$, an integer of about 2.43 million bits, is positive. The successive ratios $2178 \to 81935 \to 3082582$ are close to $e^{2\pi/\sqrt{3}} \approx 37.6$. This matches the indicial equation $\alpha^2 + \alpha + 1 = 0$ of the normalized recurrence below, whose solutions behave like $n^{-1/2}\cos(\frac{\sqrt{3}}{2}\log n + \phi)$. This heuristic only explains the pattern; the disproof is the exact computation.

3. PROOF OF (B)

Put $b(n) = a(n)/(n!)^2$. For $n \geq 1$,

$$(n+1)^2 b(n+1) = 2n(n+1)\, b(n) - n^2\, b(n-1), \qquad b(0) = 1,\ b(1) = 0. \tag{1}$$

Let v be the solution of (1) with $v(0) = 0$, $v(1) = 1$.

Lemma 2. *$|v(n)| \leq n$ and $|b(n)| \leq n$ for all $n \geq 1$.*

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Proof. Let u be either solution, and put $s(n) = n\,u(n)$ and $\delta(n) = s(n) - s(n-1)$. For $n \geq 2$, (1) becomes $s(n+1) = 2s(n) - s(n-1) - \frac{s(n-1)}{n^2-1}$, that is,

$$\delta(n+1) = \delta(n) - \frac{s(n-1)}{n^2-1}.$$

We claim that $|s(k)| \leq k^2$ for $1 \leq k \leq n$ and $|\delta(k)| \leq k$ for $2 \leq k \leq n$. This holds for $n = 2$, because $(s(1), s(2))$ is $(1, 2)$ for v and $(0, -\frac{1}{2})$ for b. Inductively,

$$|\delta(n+1)| \leq n + \frac{(n-1)^2}{n^2-1} < n+1 \qquad \text{and} \qquad |s(n+1)| \leq n^2 + n + 1 \leq (n+1)^2.$$

Hence $|u(n)| = |s(n)|/n \leq n$. □

Lemma 3 (Casoratian)**.** *$b(n)v(n-1) - b(n-1)v(n) = -1/n^2$ for all $n \geq 1$.*

Proof. The value at $n = 1$ is -1. Multiply (1) for b by $v(n)$, multiply (1) for v by $b(n)$, and subtract. This gives $(n+1)^2 W(n+1) = n^2 W(n)$, where $W(n) = b(n)v(n-1) - b(n-1)v(n)$. □

Proof of (b). By the two lemmas, for $n \geq 2$,

$$\frac{1}{n^2} \leq |b(n)|(n-1) + |b(n-1)|\,n \leq (2n-1)\max\big(|b(n)|, |b(n-1)|\big).$$

So for each $n \geq 2$ some $m_n \in \{n-1, n\}$ has $|b(m_n)| \geq 1/(2n^3)$. The set M of these m_n is infinite. For $m \in M$, using $m! \geq (m/e)^m$,

$$\frac{|a(m)|^{1/m}}{m^2} = \frac{(m!)^{2/m}\,|b(m)|^{1/m}}{m^2} \;\geq\; e^{-2}\big(2(m+1)^3\big)^{-1/m} \;\longrightarrow\; e^{-2}.$$

Conversely, $|b(n)| \leq n$ and $(n!)^{2/n} \leq (n+1)^2/e^2 \cdot (e\,(n+1))^{2/n}$ give $\limsup \leq e^{-2}$. □

4. Formalization and checks

The negation of `OeisA217703.conjecture3` is formalized in Lean 4 with Mathlib as `OeisA217703.conjecture3_false`. Its type is literally ¬ (the repository statement), which is checked by `rfl`. The formal proof works with integers. It uses the Casoratian identity $a(k+1)w(k) - a(k)w(k+1) = -(k!)^4$ for the second solution $w = (n!)^2 v$, a crude bound $|w(n)| \leq 3^n (n!)^2$, and $n^n \leq 3^n n!$. From these it shows that $|a(n)|^{1/n}/n^2 \geq 1/108$ for infinitely many n. The proof depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. Part (a) is a pure computation with numbers of millions of bits, and we have not formalized it. Our scripts check Lemmas 2 and 3 with exact rationals for $n < 3000$ and compute the signs of $a(n)$ exactly for $n \leq 81940$.

References

[1] Google DeepMind, *Formal Conjectures*, https://github.com/google-deepmind/formal-conjectures; the statements are in `FormalConjectures/OEIS/217703.lean`, pull request 6426 (2026).

[2] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org/A217703.